Advanced Database Management Systems (ADBMS)

Assignment 0 & Assignment 1: Histogram Analysis Report

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1 Advanced Database Management Systems (ADBMS)

1.1 Assignment 0 & Assignment 1: Histogram Analysis Report

2 Assignment 0

2.1 Setup

- **CPU Model & Specification**: 13th Gen Intel(R) Core(TM) i3-1315U (6 cores, 8 threads)
- **RAM Capacity**: 14 GiB
- **HDD / Storage Capacity**: 456 GiB NVMe SSD
- **Operating System**: Ubuntu 26.04 LTS
- **PostgreSQL Engine Version**: `PostgreSQL 18.6 (Ubuntu 18.6-0ubuntu0.26.04.1) on x86_64-pc-lin compiled by gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, 64-bit`

2.2 Database

- **Database Name**: Join Order Benchmark (2013 Snapshot)
- **Total Tables**: 21
- **Total Relational Tuples**: 74,190,187

2.2.1 Table Cardinalities & Primary Keys

Table Name	Primary Key Column	Cardinality (`count(*)`)
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

3 Assignment 1: Histogram

3.1 Q1. Histogram Analysis for `id` and `title` Columns of `title` Table

3.1.1 1. Corresponding SQL Query

to see what all colums needed form 'pg_stats'

```
\d pg_stats;
```

to see histogram boundary values and other relevent stored in `pg_stats`:

```
SELECT
    attname,
    null_frac,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
  AND attname IN ('id', 'title');
```

3.1.2 2. Kind of Histogram

PSQL uses an Equi Depth Histogram combined with a Most Common Values list. In an equidepth histogram, all buckets contain approximately the same number of rows ($approx N/B$), while bucket boundaries vary in width according to data distribution

3.1.3 3. Number of Buckets

- Column `id`: **100 buckets** (101 bucket boundary values)
- Column `title`: **100 buckets** (101 bucket boundary values)
- extra - to see the defualt number of buckets

```
SHOW default_statistics_target;
```

3.1.4 4. How Textual Attribute `title` is Put into Histogram

textual attribute "title" is put into histogram as:

1. sorting the text strings in lexicographical order like bannana and apple will be apple, bannana after sorting

2. Sampling string values from the table like table title has around 2.5M rows , it will only sample around 30K

3. Encoding initial string prefix bytes into floating-point numbers in $[0.0, 1.0]$ range for quantitative interpolation like laptop will be broken to lap with l = 108, a=97, p=112 then by 256 as between 0 and 1 , lap will be around 0.42 , we can then see where a key belongs between two keeys

4. Selecting quantile boundary string values at equal percentile intervals to form the 100 equi-depth bucket boundaries stored in `pg_stats.histogram_bounds`.

3.1.5 5. Exact Bucket Boundaries List (From `pg_stats`)

A. Column `id` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`id`)	Bucket Interval
B000 (Start)	41	Start of table data
B001	25598	Bucket 1: [41 - 25598]
B002	50067	Bucket 2: [25598 - 50067]
B003	75257	Bucket 3: [50067 - 75257]
B004	101946	Bucket 4: [75257 - 101946]
B005	129223	Bucket 5: [101946 - 129223]
B006	155973	Bucket 6: [129223 - 155973]
B007	181990	Bucket 7: [155973 - 181990]
B008	208153	Bucket 8: [181990 - 208153]
B009	234525	Bucket 9: [208153 - 234525]

B010	259507	Bucket 10: [234525 - 259507]
B011	287845	Bucket 11: [259507 - 287845]
B012	314618	Bucket 12: [287845 - 314618]
B013	340479	Bucket 13: [314618 - 340479]
B014	366364	Bucket 14: [340479 - 366364]
B015	393076	Bucket 15: [366364 - 393076]
B016	418929	Bucket 16: [393076 - 418929]
B017	443204	Bucket 17: [418929 - 443204]
B018	472539	Bucket 18: [443204 - 472539]
B019	497095	Bucket 19: [472539 - 497095]
B020	521015	Bucket 20: [497095 - 521015]
B021	547008	Bucket 21: [521015 - 547008]
B022	571914	Bucket 22: [547008 - 571914]
B023	600124	Bucket 23: [571914 - 600124]
B024	624290	Bucket 24: [600124 - 624290]
B025	648346	Bucket 25: [624290 - 648346]
B026	673077	Bucket 26: [648346 - 673077]
B027	698192	Bucket 27: [673077 - 698192]
B028	723945	Bucket 28: [698192 - 723945]
B029	747971	Bucket 29: [723945 - 747971]
B030	774715	Bucket 30: [747971 - 774715]
B031	801886	Bucket 31: [774715 - 801886]
B032	825206	Bucket 32: [801886 - 825206]
B033	852044	Bucket 33: [825206 - 852044]
B034	878630	Bucket 34: [852044 - 878630]
B035	903227	Bucket 35: [878630 - 903227]
B036	929542	Bucket 36: [903227 - 929542]
B037	951992	Bucket 37: [929542 - 951992]
B038	976177	Bucket 38: [951992 - 976177]
B039	1002223	Bucket 39: [976177 - 1002223]
B040	1027496	Bucket 40: [1002223 - 1027496]
B041	1053013	Bucket 41: [1027496 - 1053013]
B042	1076669	Bucket 42: [1053013 - 1076669]
B043	1101644	Bucket 43: [1076669 - 1101644]
B044	1127312	Bucket 44: [1101644 - 1127312]
B045	1153408	Bucket 45: [1127312 - 1153408]
B046	1177087	Bucket 46: [1153408 - 1177087]
B047	1200809	Bucket 47: [1177087 - 1200809]
B048	1228132	Bucket 48: [1200809 - 1228132]
B049	1251987	Bucket 49: [1228132 - 1251987]
B050	1276194	Bucket 50: [1251987 - 1276194]
B051	1300208	Bucket 51: [1276194 - 1300208]
B052	1325279	Bucket 52: [1300208 - 1325279]
B053	1350079	Bucket 53: [1325279 - 1350079]
B054	1375217	Bucket 54: [1350079 - 1375217]
B055	1399305	Bucket 55: [1375217 - 1399305]
B056	1424915	Bucket 56: [1399305 - 1424915]
B057	1447763	Bucket 57: [1424915 - 1447763]
B058	1472187	Bucket 58: [1447763 - 1472187]
B059	1496066	Bucket 59: [1472187 - 1496066]
B060	1520933	Bucket 60: [1496066 - 1520933]
B061	1548374	Bucket 61: [1520933 - 1548374]
B062	1572984	Bucket 62: [1548374 - 1572984]
B063	1597331	Bucket 63: [1572984 - 1597331]
B064	1621524	Bucket 64: [1597331 - 1621524]
B065	1646529	Bucket 65: [1621524 - 1646529]
B066	1671841	Bucket 66: [1646529 - 1671841]
B067	1697305	Bucket 67: [1671841 - 1697305]
B068	1720750	Bucket 68: [1697305 - 1720750]

B069	1746578	Bucket 69: [1720750 - 1746578]
B070	1771397	Bucket 70: [1746578 - 1771397]
B071	1796312	Bucket 71: [1771397 - 1796312]
B072	1821069	Bucket 72: [1796312 - 1821069]
B073	1846620	Bucket 73: [1821069 - 1846620]
B074	1871455	Bucket 74: [1846620 - 1871455]
B075	1897069	Bucket 75: [1871455 - 1897069]
B076	1923039	Bucket 76: [1897069 - 1923039]
B077	1951619	Bucket 77: [1923039 - 1951619]
B078	1975550	Bucket 78: [1951619 - 1975550]
B079	2001398	Bucket 79: [1975550 - 2001398]
B080	2025488	Bucket 80: [2001398 - 2025488]
B081	2048645	Bucket 81: [2025488 - 2048645]
B082	2075506	Bucket 82: [2048645 - 2075506]
B083	2099704	Bucket 83: [2075506 - 2099704]
B084	2124027	Bucket 84: [2099704 - 2124027]
B085	2146547	Bucket 85: [2124027 - 2146547]
B086	2170970	Bucket 86: [2146547 - 2170970]
B087	2195637	Bucket 87: [2170970 - 2195637]
B088	2221366	Bucket 88: [2195637 - 2221366]
B089	2249864	Bucket 89: [2221366 - 2249864]
B090	2274023	Bucket 90: [2249864 - 2274023]
B091	2299152	Bucket 91: [2274023 - 2299152]
B092	2325178	Bucket 92: [2299152 - 2325178]
B093	2352562	Bucket 93: [2325178 - 2352562]
B094	2378000	Bucket 94: [2352562 - 2378000]
B095	2403761	Bucket 95: [2378000 - 2403761]
B096	2428038	Bucket 96: [2403761 - 2428038]
B097	2452338	Bucket 97: [2428038 - 2452338]
B098	2479928	Bucket 98: [2452338 - 2479928]
B099	2505988	Bucket 99: [2479928 - 2505988]
B100	2528305	Bucket 100: [2505988 - 2528305]

B. Column `title` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`title`)	Lexicographical Interval
B000 (Start)	...	Start of text data
B001	(#1.1302)	Bucket 1: [... - (#1.1302)]
B002	(#1.219)	Bucket 2: [(#1.1302) - (#1.219)]
B003	(#1.3662)	Bucket 3: [(#1.219) - (#1.3662)]
B004	(#1.59)	Bucket 4: [(#1.3662) - (#1.59)]
B005	(#1.846)	Bucket 5: [(#1.59) - (#1.846)]
B006	(1968-02-14)	Bucket 6: [(#1.846) - (1968-02-14)]
B007	(1982-12-14)	Bucket 7: [(1968-02-14) - (1982-12-14)]
B008	(1994-07-07)	Bucket 8: [(1982-12-14) - (1994-07-07)]
B009	(1998-03-16)	Bucket 9: [(1994-07-07) - (1998-03-16)]
B010	(2000-01-22)	Bucket 10: [(1998-03-16) - (2000-01-22)]
B011	(2001-11-06)	Bucket 11: [(2000-01-22) - (2001-11-06)]
B012	(2003-08-29)	Bucket 12: [(2001-11-06) - (2003-08-29)]
B013	(2005-01-19)	Bucket 13: [(2003-08-29) - (2005-01-19)]
B014	(2006-05-17)	Bucket 14: [(2005-01-19) - (2006-05-17)]
B015	(2007-07-05)	Bucket 15: [(2006-05-17) - (2007-07-05)]

B016	(2009-02-02)	Bucket 16: [(2007-07-05) - (2009-02-02)]
B017	(2010-09-07)	Bucket 17: [(2009-02-02) - (2010-09-07)]
B018	(2011-08-29)	Bucket 18: [(2010-09-07) - (2011-08-29)]
B019	(2012-07-21)	Bucket 19: [(2011-08-29) - (2012-07-21)]
B020	(#20.140)	Bucket 20: [(2012-07-21) - (#20.140)]
B021	(#2.92)	Bucket 21: [(#20.140) - (#2.92)]
B022	(#4.172)	Bucket 22: [(#2.92) - (#4.172)]
B023	(#6.163)	Bucket 23: [(#4.172) - (#6.163)]
B024	Aamu joutuu pesemään tekojensa jälkipyykkiä	Bucket 24: [(#6.163) - Aamu joutuu pesemään tekojensa jälkipyykkiä]
B025	Aflevering 2559	Bucket 25: [Aamu joutuu pesemään tekojensa jälkipyykkiä - Aflevering 2559]
B026	A Little Trip	Bucket 26: [Aflevering 2559 - A Little Trip]
B027	An Atheist Named Faith	Bucket 27: [A Little Trip - An Atheist Named Faith]
B028	Arabeskebi Pirosmanis temaze	Bucket 28: [An Atheist Named Faith - Arabeskebi Pirosmanis temaze]
B029	Attenborough's Ark: Natural World Special	Bucket 29: [Arabeskebi Pirosmanis temaze - Attenborough's Ark: Natural World Special]
B030	Baldur Blauzahn	Bucket 30: [Attenborough's Ark: Natural World Special - Baldur Blauzahn]
B031	Best in Show Cakes	Bucket 31: [Baldur Blauzahn - Best in Show Cakes]
B032	Blues	Bucket 32: [Best in Show Cakes - Blues]
B033	Brown Before Green	Bucket 33: [Blues - Brown Before Green]
B034	Captive Audience	Bucket 34: [Brown Before Green - Captive Audience]
B035	Cheryl Cole's Night In	Bucket 35: [Captive Audience - Cheryl Cole's Night In]
B036	Cobra Starship	Bucket 36: [Cheryl Cole's Night In - Cobra Starship]
B037	CPM 22 e Babado Novo	Bucket 37: [Cobra Starship - CPM 22 e Babado Novo]
B038	Dark Moon	Bucket 38: [CPM 22 e Babado Novo - Dark Moon]
B039	De intrede van de Sint - Antwerpen 2011	Bucket 39: [Dark Moon - De intrede van de Sint - Antwerpen 2011]
B040	Det' politiet: Part 2	Bucket 40: [De intrede van de Sint - Antwerpen 2011 - Det' politiet: Part 2]
B041	DIY Network Holiday Celebration	Bucket 41: [Det' politiet: Part 2 - DIY Network Holiday Celebration]
B042	Drugog puta nema!	Bucket 42: [DIY Network Holiday Celebration - Drugog puta nema!]
B043	Elements	Bucket 43: [Drugog puta nema! - Elements]
B044	Episode V - Tag der Abrechnung	Bucket 44: [Elements - Episode V - Tag der Abrechnung]
B045	Famiresu tenshi purin	Bucket 45: [Episode V - Tag der Abrechnung - Famiresu tenshi purin]

B046	Fleshdance	Bucket 46: [Famiresu tenshi purin - Fleshdance]
B047	Fuga y misterio	Bucket 47: [Fleshdance - Fuga y misterio]
B048	Girls' Generation and the Dangerous Boys	Bucket 48: [Fuga y misterio - Girls' Generation and the Dangerous Boys]
B049	Guadalcanal/Leckie	Bucket 49: [Girls' Generation and the Dangerous Boys - Guadalcanal/Leckie]
B050	Heaven on Earth	Bucket 50: [Guadalcanal/Leckie - Heaven on Earth]
B051	Holiday	Bucket 51: [Heaven on Earth - Holiday]
B052	Hurricane Katrina Aftermath	Bucket 52: [Holiday - Hurricane Katrina Aftermath]
B053	Inauguração na Estrela - Lisboa	Bucket 53: [Hurricane Katrina Aftermath - Inauguração na Estrela - Lisboa]
B054	Is There Still Hope for Gino and Mikay?	Bucket 54: [Inauguração na Estrela - Lisboa - Is There Still Hope for Gino and Mikay?]
B055	Jindalee Lady	Bucket 55: [Is There Still Hope for Gino and Mikay? - Jindalee Lady]
B056	Karina	Bucket 56: [Jindalee Lady - Karina]
B057	Know Thy Partner	Bucket 57: [Karina - Know Thy Partner]
B058	Ladrones de niños	Bucket 58: [Know Thy Partner - Ladrones de niños]
B059	Late as a Motherer	Bucket 59: [Ladrones de niños - Late as a Motherer]
B060	Les éperlans	Bucket 60: [Late as a Motherer - Les éperlans]
B061	Living Room Upgrade	Bucket 61: [Les éperlans - Living Room Upgrade]
B062	Luha ng ina	Bucket 62: [Living Room Upgrade - Luha ng ina]
B063	Männer sind auch nur Menschen	Bucket 63: [Luha ng ina - Männer sind auch nur Menschen]
B064	Melissa	Bucket 64: [Männer sind auch nur Menschen - Melissa]
B065	Mission to Africa	Bucket 65: [Melissa - Mission to Africa]
B066	Mr. Deity and the Days	Bucket 66: [Mission to Africa - Mr. Deity and the Days]
B067	Nanna & Lil' Puss Puss in 'The Honeymoonies'	Bucket 67: [Mr. Deity and the Days - Nanna & Lil' Puss Puss in 'The Honeymoonies']
B068	Ninna Hyttinen	Bucket 68: [Nanna & Lil' Puss Puss in 'The Honeymoonies' - Ninna Hyttinen]
B069	Odoriko no yubiwa	Bucket 69: [Ninna Hyttinen - Odoriko no yubiwa]
B070	Operation: Bank Job	Bucket 70: [Odoriko no yubiwa - Operation: Bank Job]
B071	Papa wo sagashite!	Bucket 71: [Operation: Bank Job - Papa wo sagashite!]
B072	Phantom Fever	Bucket 72: [Papa wo sagashite! - Phantom Fever]
B073	Power of the South	Bucket 73: [Phantom Fever - Power of the South]
B074	Quatre acteurs à bout de souffle	Bucket 74: [Power of the South - Quatre acteurs à bout de souffle]

B075	Relative Air Pollution	Bucket 75: [Quatre acteurs à bout de souffle - Relative Air Pollution]
B076	Rolling Stones	Bucket 76: [Relative Air Pollution - Rolling Stones]
B077	San Diego, Bukarest, Paris	Bucket 77: [Rolling Stones - San Diego, Bukarest, Paris]
B078	Seekers	Bucket 78: [San Diego, Bukarest, Paris - Seekers]
B079	Shônen Tantei to Supai	Bucket 79: [Seekers - Shônen Tantei to Supai]
B080	Smash Hit: She	Bucket 80: [Shônen Tantei to Supai - Smash Hit: She]
B081	Spilling the Beans	Bucket 81: [Smash Hit: She - Spilling the Beans]
B082	Strider 2	Bucket 82: [Spilling the Beans - Strider 2]
B083	Take This Job and Shove It	Bucket 83: [Strider 2 - Take This Job and Shove It]
B084	The Accident	Bucket 84: [Take This Job and Shove It - The Accident]
B085	The Captain's Son: Untitled	Bucket 85: [The Accident - The Captain's Son: Untitled]
B086	The End of the World as We Know It	Bucket 86: [The Captain's Son: Untitled - The End of the World as We Know It]
B087	The Healers	Bucket 87: [The End of the World as We Know It - The Healers]
B088	The Lobsters Cook	Bucket 88: [The Healers - The Lobsters Cook]
B089	The Odyssey	Bucket 89: [The Lobsters Cook - The Odyssey]
B090	The Ring of Ultimate Power	Bucket 90: [The Odyssey - The Ring of Ultimate Power]
B091	The Thing in Mrs. Faversham's Attic	Bucket 91: [The Ring of Ultimate Power - The Thing in Mrs. Faversham's Attic]
B092	Thin Ice	Bucket 92: [The Thing in Mrs. Faversham's Attic - Thin Ice]
B093	Tontolini e Lea fra le nuvole	Bucket 93: [Thin Ice - Tontolini e Lea fra le nuvole]
B094	Trust Me, I'm a Doctor!	Bucket 94: [Tontolini e Lea fra le nuvole - Trust Me, I'm a Doctor!]
B095	Undertryk	Bucket 95: [Trust Me, I'm a Doctor! - Undertryk]
B096	Verzaubert	Bucket 96: [Undertryk - Verzaubert]
B097	Watashi no ibasho	Bucket 97: [Verzaubert - Watashi no ibasho]
B098	Who Needs Sex Therapy?	Bucket 98: [Watashi no ibasho - Who Needs Sex Therapy?]
B099	Yasmina	Bucket 99: [Who Needs Sex Therapy? - Yasmina]
B100	Zzzzzz	Bucket 100: [Yasmina - Zzzzzz]

3.1.6 6. Histogram Visualization

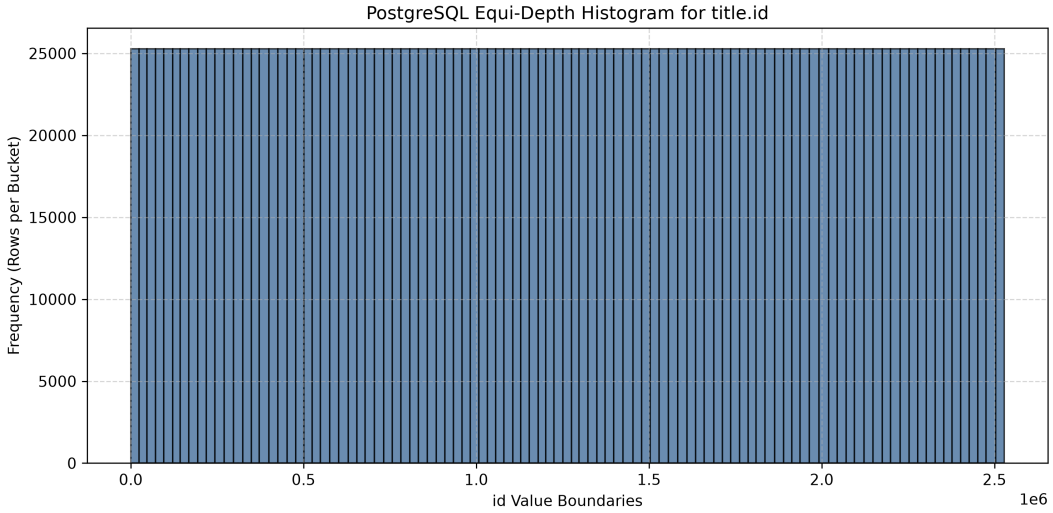


Figure 1: Histogram for title.id

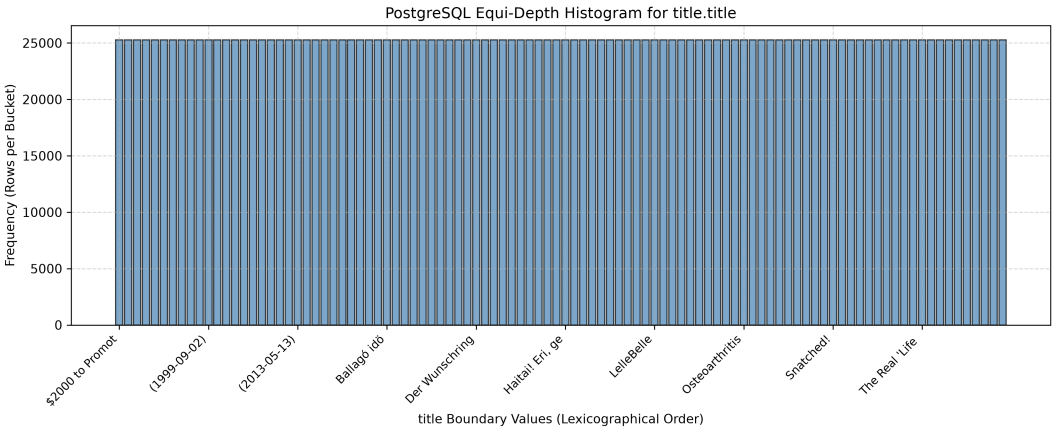


Figure 2: Histogram for title.title

3.1.7 7. Max Selectivity Error

- **Max Selectivity Error**: Since `default_statistics_target` is 100, there are 100 buckets. Therefore, the maximum selectivity error for a range query is bounded by:

$$\text{Max Error} \leq \frac{1}{B} = \frac{1}{100} = 0.01 \quad (1.0\%)$$

For evaluated queries(q9), the observed maximum selectivity error is **0.8364%** (0.008364), which strictly satisfies Max Error $\leq 1.0\%$.

3.1.8 8: Selectivity Estimation for Text Columns

Selectivity estimate is done for range query on text column such as "title" in PSQL such as:

*The engine scans the `histogram_bounds` array stored under pg_stats to locate the specific bucket where the target string is, after that it counts all complete buckets preceding the target bucket. Each full bucket represents a fixed fraction of total rows (1/B). For the target bucket, PostgreSQL converts initial string bytes into floating-point numbers in $[0.0, 1.0]$ range for interpolation as explain in part 4, assuming uniform distribution inside the target bucket, it calculates the target's position using:

$$\text{Fraction} = \frac{\text{Query Value} - \text{Bucket Min}}{\text{Bucket Max} - \text{Bucket Min}}$$

- Final Calculation is done by adding the fractional bucket estimate to the sum of preceding full buckets to get the final estimated selectivity.

3.1.9 9: Estimated vs. Actual Selectivities

To calculate estimated and actual selectivities of the queries and moreover there accuracy we compare estimated rows (from `EXPLAIN`) against actual rows (from `EXPLAIN ANALYZE`).

Total row count (N) for the `title` table is 2,528,312.

Query 1: `title < 'Race'`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.title < 'Race';
```

- **Actual Selectivity:** The query physically returned 1,918,141 matching rows: $S_{\text{act}} = \frac{1918141}{2528312} \approx 0.7587$ (75.87%).

```
EXPLAIN SELECT * FROM title WHERE title.title < 'Race';
```

- **Estimated Selectivity:** The planner estimated 1,939,288 rows: $S_{\text{est}} = \frac{1939288}{2528312} \approx 0.7670$ (76.70%).
- **Observation:** The absolute error is roughly 0.83%.

Query 2: `id < 350000`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.id < '350000';
```

- **Actual Selectivity:** The query returned 349,999 matching rows: $S_{\text{act}} = \frac{349999}{2528312} \approx 0.1384$ (13.84%).

```
EXPLAIN SELECT * FROM title WHERE title.id < '350000';
```

- **Estimated Selectivity:** The planner estimated 337,982 rows: $S_{\text{est}} = \frac{337982}{2528312} \approx 0.1337$ (13.37%).
- **Observation:** The absolute error here is about 0.47%.

3.2 Q2. Optimal Serial Histogram Construction ($B = 20$ Buckets)

3.2.1 1. Histogram Visualizations for $N = 3000$ Samples (Horizontal Bar Plots)

As specified in ques the histograms as a bar-plot with value boundaries in the Y-axis and frequency in the X-axis

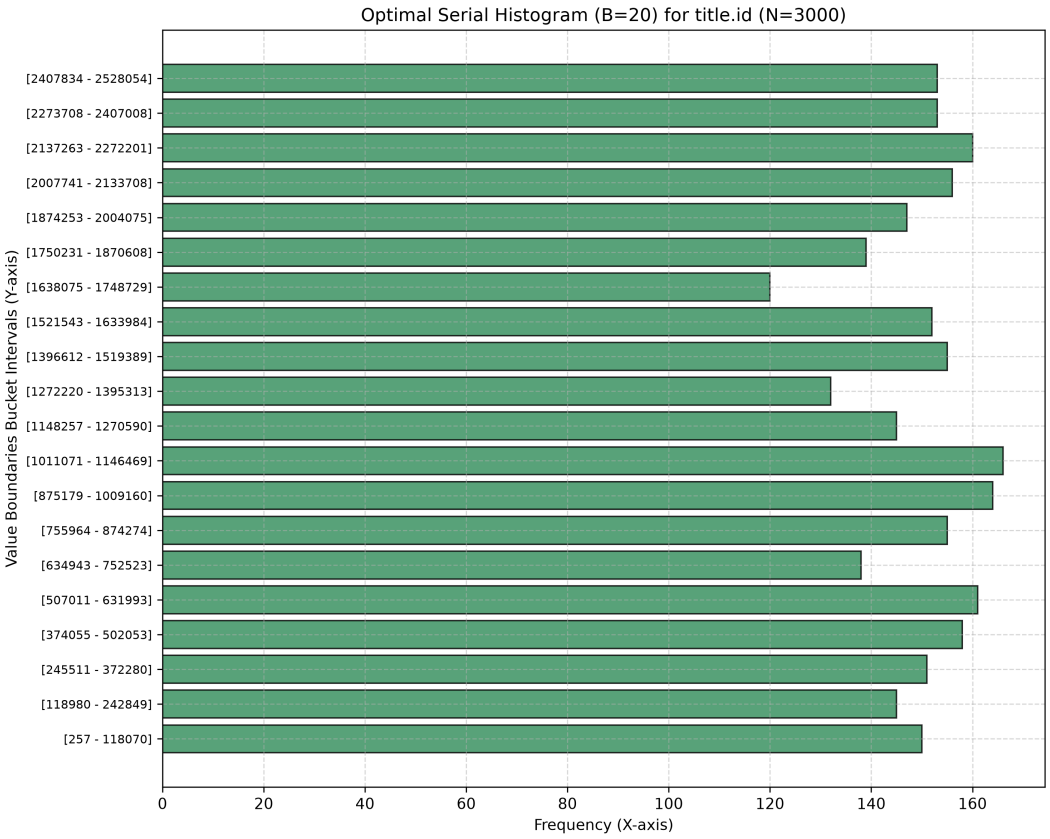


Figure 3: Optimal Serial Histogram for title.id (N=3000)

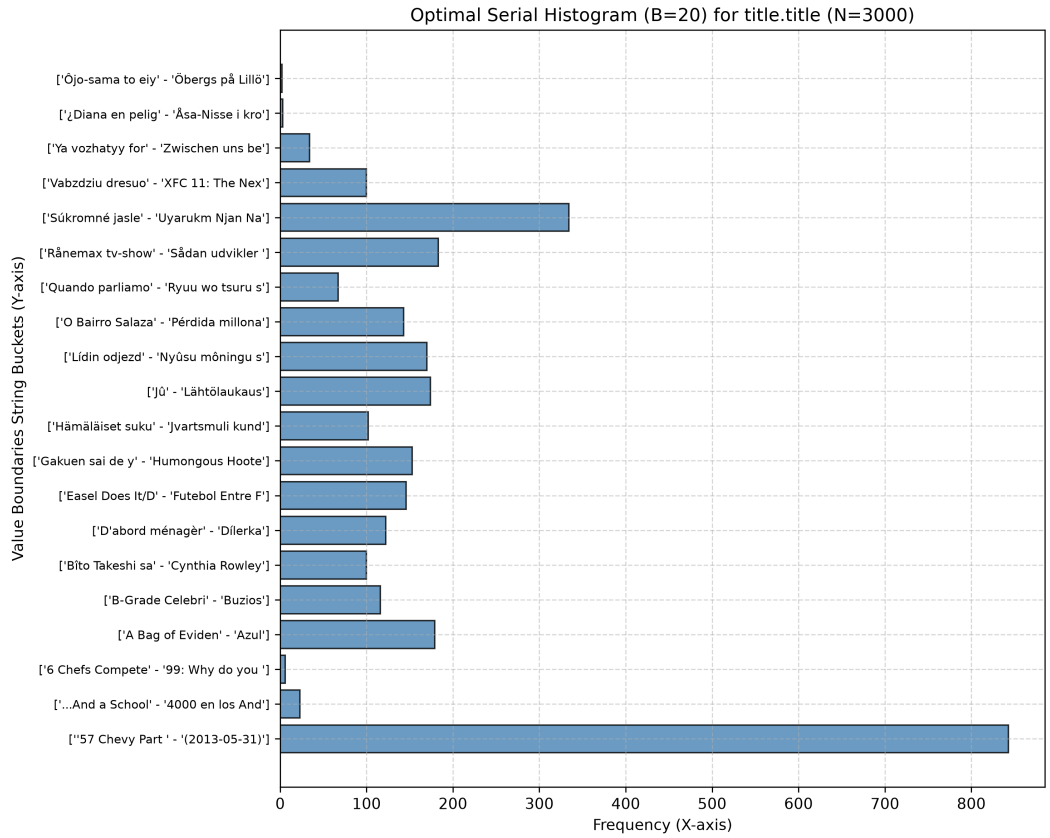


Figure 4: Optimal Serial Histogram for title.title (N=3000)

3.2.2 2. Execution Build Time

- **Execution Build Time for $N = 3000$: 0.8290 seconds **(`id` column)** / 0.8240 seconds** (`title` column)

3.2.3 3. Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

The optimal serial histogram algorithm partitions the 3,000 samples into 20 dynamic buckets to minimize total Sum of Squared Errors . The exact bucket intervals and row frequency counts (n_i) for both columns are listed below:

A. Column `id` Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

Bucket #	**Value Interval (`id`)**	**Frequency Count (n_i)**
Bucket 01	[257 - 118,070]	150
Bucket 02	[118,980 - 242,849]	145
Bucket 03	[245,511 - 372,280]	151
Bucket 04	[374,055 - 502,053]	158
Bucket 05	[507,011 - 631,993]	161
Bucket 06	[634,943 - 752,523]	138
Bucket 07	[755,964 - 874,274]	155
Bucket 08	[875,179 - 1,009,160]	164
Bucket 09	[1,011,071 - 1,146,469]	166
Bucket 10	[1,148,257 - 1,270,590]	145
Bucket 11	[1,272,220 - 1,395,313]	132
Bucket 12	[1,396,612 - 1,519,389]	155
Bucket 13	[1,521,543 - 1,633,984]	152
Bucket 14	[1,638,075 - 1,748,729]	120
Bucket 15	[1,750,231 - 1,870,608]	139
Bucket 16	[1,874,253 - 2,004,075]	147
Bucket 17	[2,007,741 - 2,133,708]	156
Bucket 18	[2,137,263 - 2,272,201]	160
Bucket 19	[2,273,708 - 2,407,008]	153
Bucket 20	[2,407,834 - 2,528,054]	153

B. Column `title` Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

Bucket #	**Value Interval (`title`)**	**Frequency Count (n_i)**
Bucket 01	["'57 Chevy Part 2" - "(2013-05-31)"]	843
Bucket 02	["...And a School Bus" - "4000 en los Andes"]	23
Bucket 03	["6 Chefs Compete" - "99: Why do you say that?"]	6
Bucket 04	["A Bag of Evidence" - "Azul"]	179
Bucket 05	["B-Grade Celebrity/Bucks P" - "Buzios"]	116
Bucket 06	["Bîto Takeshi satsujin jik" - "Cynthia Rowley"]	100
Bucket 07	["D'abord ménagères" - "Dílerka"]	122
Bucket 08	["Easel Does It/Dinky at th" - "Futebol Entre Fluminense "]	146
Bucket 09	["Gakuen sai de yui, hansei" - "Humongous Hooters"]	153
Bucket 10	["Hämäläiset sukupidot Tamm" - "Jvartsmuli kundzuli"]	102
Bucket 11	["Jû" - "Lähtölaukaus"]	174
Bucket 12	["Lídin odjezd" - "Nyûsu môningu sateraito"]	170

Bucket 13	["O Bairro Salazar" - "Pérdida millonaria"]	143
Bucket 14	["Quando parliamo d'amore" - "Ryuu wo tsuru shounen"]	67
Bucket 15	["Rånemax tv-show" - "Sådan udvikler du dit bar"]	183
Bucket 16	["Súkromné jasle" - "Uyarukm Njan Nadaake"]	334
Bucket 17	["Vabzdziu dresuotojas" - "XFC 11: The Next Generati"]	100
Bucket 18	["Ya vozhatyy forposta" - "Zwischen uns beiden"]	34
Bucket 19	["¿Diana en peligro?" - "Åsa-Nisse i kronans kläde"]	3
Bucket 20	["Ôjo-sama to eiyû" - "Öbergs på Lillöga"]	2

3.2.4 4. Max Selectivity Error for Optimal Histogram ($N = 3000$)

- **Max Selectivity Error**: **0.003720** (or **0.3720%**)

Theoretical error , when worst case senario occurs ie it is skewed heavily

$$\text{Max Error (Theoretical)} = \frac{\max(C_i)}{N}$$

A. Step-by-Step Calculation Methodology 1. **Evaluation Quantiles ($q_0 \dots q_9$)**: To systematically evaluate estimation error across the data range $[x_{\min}, x_{\max}]$, 10 linearly-spaced query thresholds q_k are evaluated:

$$q_k = x_{\min} + k \times \frac{x_{\max} - x_{\min}}{9}, \quad k \in \{0, 1, \dots, 9\}$$

2. **Actual Selectivity ($S_{\text{act}}(q_k)$)**: Computed directly from the empirical sample of $N = 3,000$ points:

$$S_{\text{act}}(q_k) = \frac{\text{Count}(x \le q_k)}{N}$$

3. **Optimal Histogram Estimated Selectivity ($S_{\text{est}}(q_k)$)**:

- For all buckets $[L_i, H_i]$ lying fully below q_k ($H_i \le q_k$), add full bucket count n_i.
- For the enclosing bucket ($L_i \le q_k < H_i$), perform linear interpolation:

$$\text{Estimated Count } E(q_k) = \sum_{H_i \le q_k} n_i + \left(n_i \times \frac{q_k - L_i}{H_i - L_i} \right)$$

- Estimated Selectivity: $S_{\text{est}}(q_k) = \frac{E(q_k)}{N}$

4. **Maximum Selectivity Error**:

$$\text{Max Selectivity Error} = \max_{k \in \{0..9\}} |S_{\text{est}}(q_k) - S_{\text{act}}(q_k)| = \mathbf{0.003720}$$

B. Quantile-by-Quantile Error Evaluation Table ($N = 3000, B = 20$)

Index (k)	Query Threshold (q_k)	Actual Count (A)	Actual Selectivity (S_{act})	Estimated Count (E)	Estimated Selectivi
q_0	257.0	1	0.000333	0.00	0.000000
q_1	281,323.3	342	0.114000	337.42	0.112473
q_2	561,988.7	668	0.222667	674.82	0.224941
q_3	842,856.0	1026	0.342667	1016.84	0.338946
q_4	1,123,722.3	1353	0.451000	1360.11	0.453371
q_5	1,404,588.7	1678	0.557667	1675.07	0.558357
q_6	1,685,455.0	2025	0.675333	2023.38	0.674461
q_7	1,966,321.3	2341	0.781667	2335.25	0.778417
q_8	2,247,187.7	2697	0.889667	2664.34	0.888114
q_9	2,528,054.0	3000	1.000000	3000.00	1.000000

3.2.5 5. Histograms Across Sample Sizes ($N = 1000$ and $N = 3000$)

Below are the optimal serial histogram visualizations ($B = 20$) for both sample sizes $N = 1000$ and $N = 3000$:

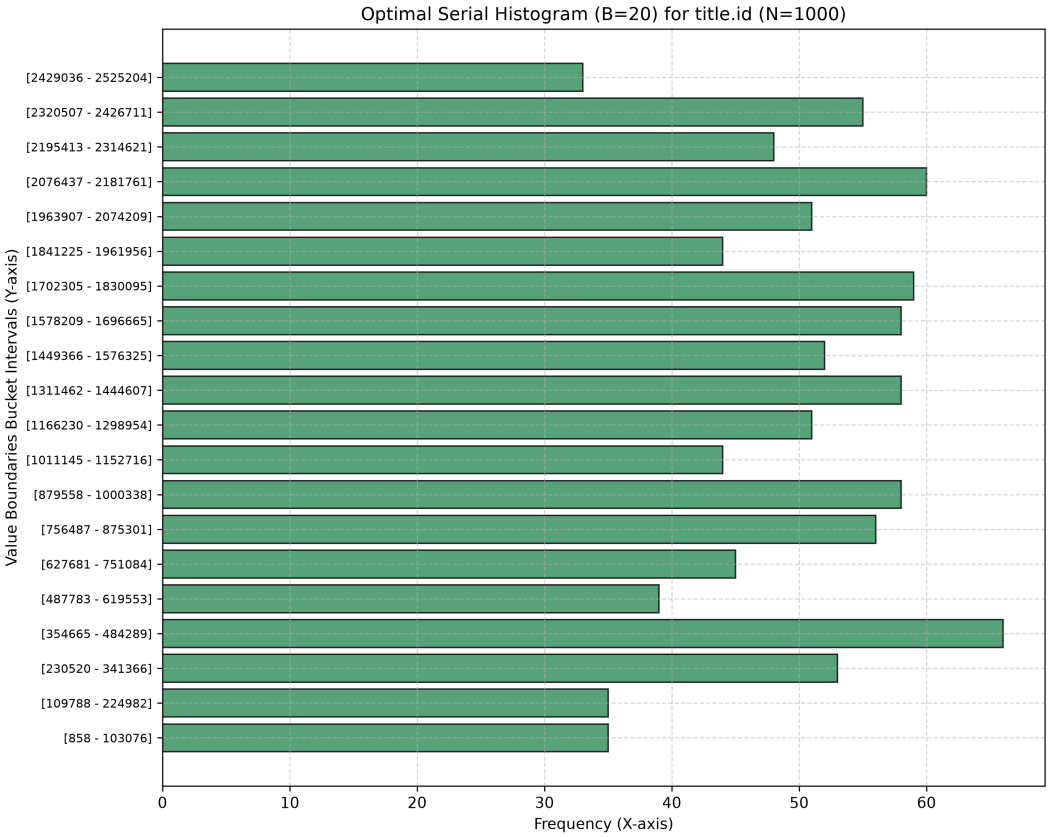


Figure 5: Optimal Serial Histogram for title.id (N=1000)

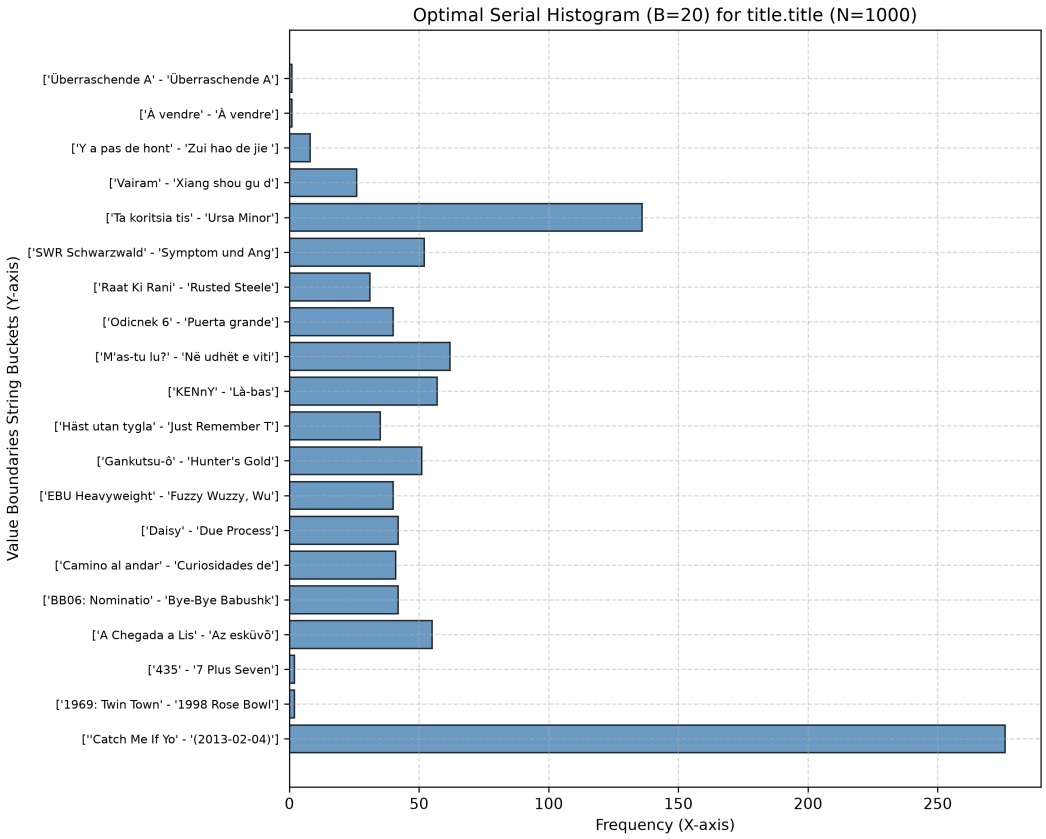


Figure 6: Optimal Serial Histogram for title.title (N=1000)

A. Sample Size $N = 1000$ Histograms

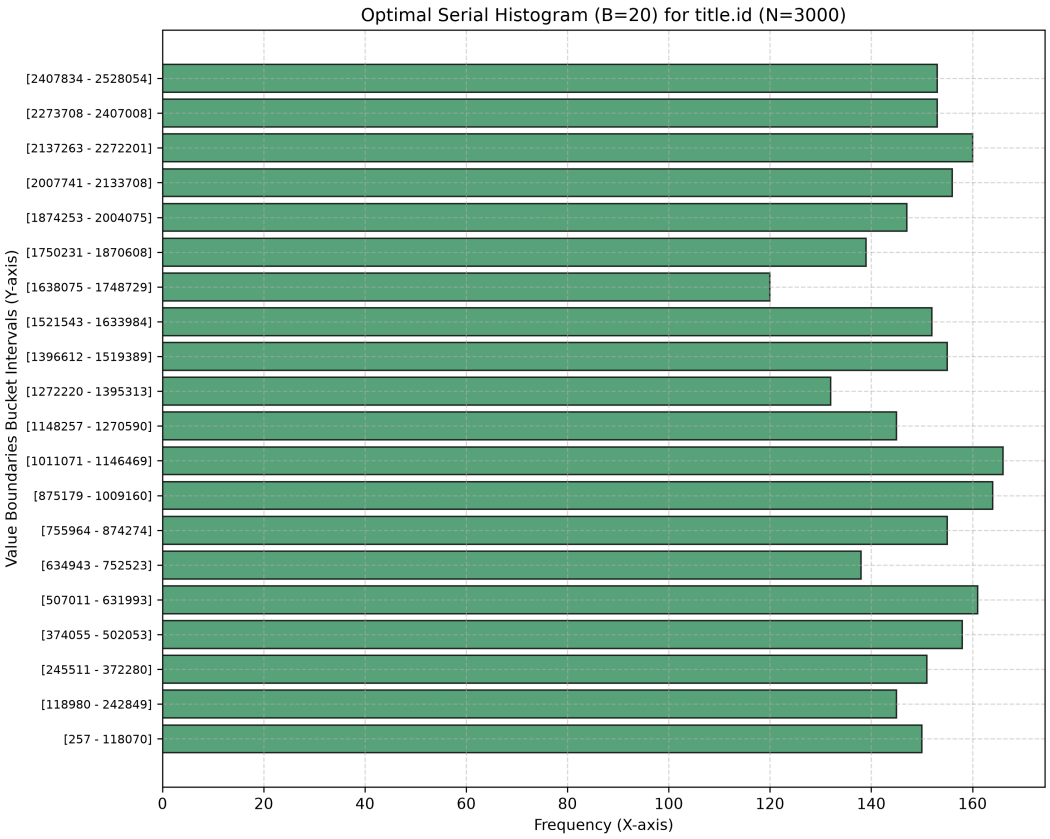


Figure 7: Optimal Serial Histogram for title.id (N=3000)

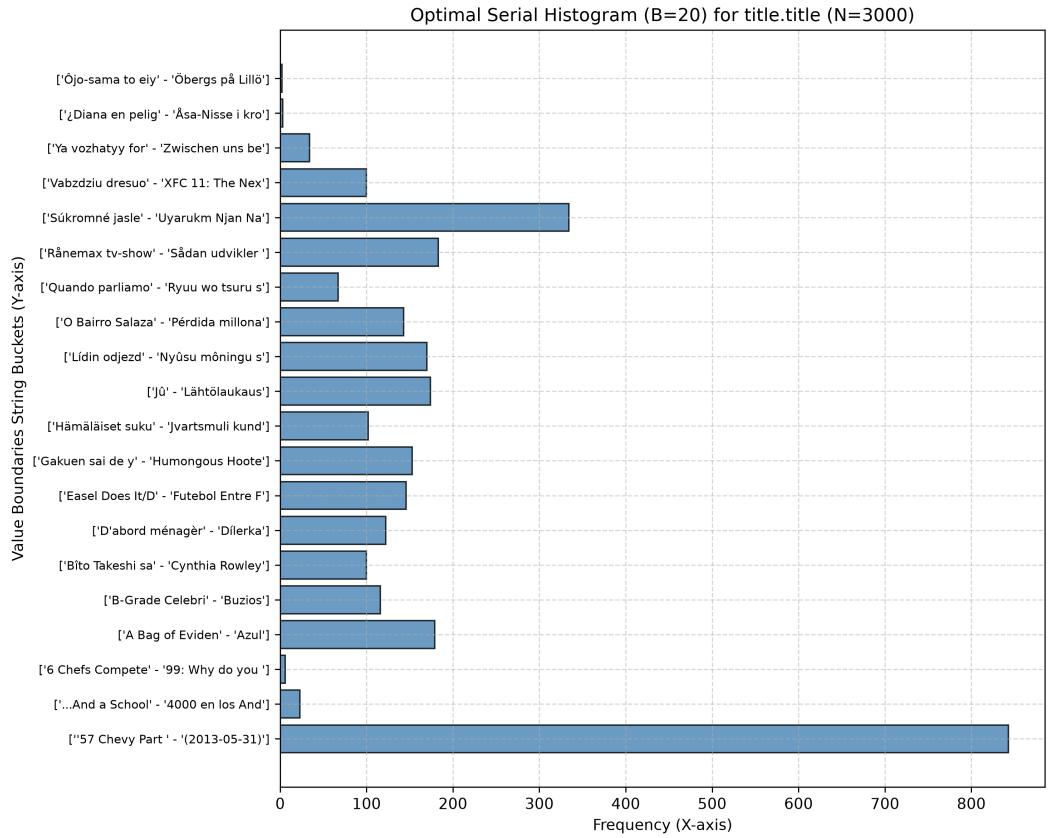


Figure 8: Optimal Serial Histogram for title.title (N=3000)

B. Sample Size $N = 3000$ Histograms

C. Summary Table Across Sample Sizes

Sample Size (N)	Build Time (`id`)	Build Time (`title`)	Max Selectivity Error
1000	0.1994s	0.1719s	0.004699
3000	0.8290s	0.8240s	0.003720
5000	2.1767s	2.0974s	0.002471

3.2.6 6. Full Step-by-Step Linear Regression Extrapolation ($N = 2,528,312$)

To extrapolate the construction build time for the entire `title` table ($N = 2,528,312$), we fit a linear regression model:

$$T(N) = m \cdot N + c$$

A. Sample Data Points (N_i, T_i)

- Point 1: $(N_1, T_1) = (1000, 0.1994)$
- Point 2: $(N_2, T_2) = (3000, 0.8290)$
- Point 3: $(N_3, T_3) = (5000, 2.1767)$

B. Mean Calculations

- $\bar{N} = \frac{1000+3000+5000}{3} = \mathbf{3000}$
- $\bar{T} = \frac{0.1994+0.8290+2.1767}{3} = \mathbf{1.0684}$ seconds

C. Deviations & Sum of Products

N_i	T_i (s)	$(N_i - \bar{N})$	$(T_i - \bar{T})$	$(N_i - \bar{N})^2$	$(N_i - \bar{N})(T_i - \bar{T})$
1000	0.1994	-2000	-0.8690	4,000,000	1738.0037
3000	0.8290	0	-0.2394	0	-0.0000
5000	2.1767	+2000	1.1084	4,000,000	2216.7320
Sum				**8,000,000**	**3954.7358**

D. Slope (m) and Intercept (c) Formulas

- **Slope (m)**:

$$m = \frac{\sum(N_i - \bar{N})(T_i - \bar{T})}{\sum(N_i - \bar{N})^2} = \frac{3954.7358}{8,000,000} = \mathbf{0.00049434} \text{ seconds / sample}$$

- **Intercept (c)**:

$$c = \bar{T} - m \cdot \bar{N} = 1.0684 - (0.00049434 \times 3000) = \mathbf{-0.414643} \text{ seconds}$$

E. Extrapolated Build Time for Full Table ($N = 2,528,312$)

$$T(2,528,312) = (0.00049434 \times 2,528,312) + (-0.414643) = \mathbf{1249.44} \text{ seconds} \quad (\mathbf{20.82} \text{ minutes})$$

3.2.7 7. Sample Size vs Time Relationship Plot

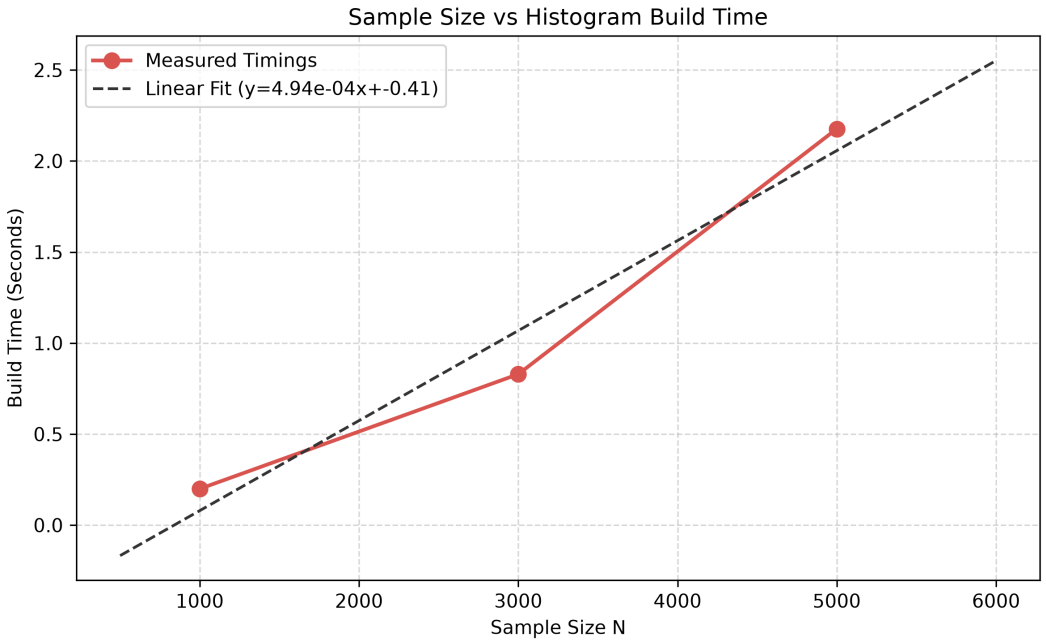


Figure 9: Sample Size vs Time

3.2.8 8. Sample Size vs Max Selectivity Error Relationship Plot

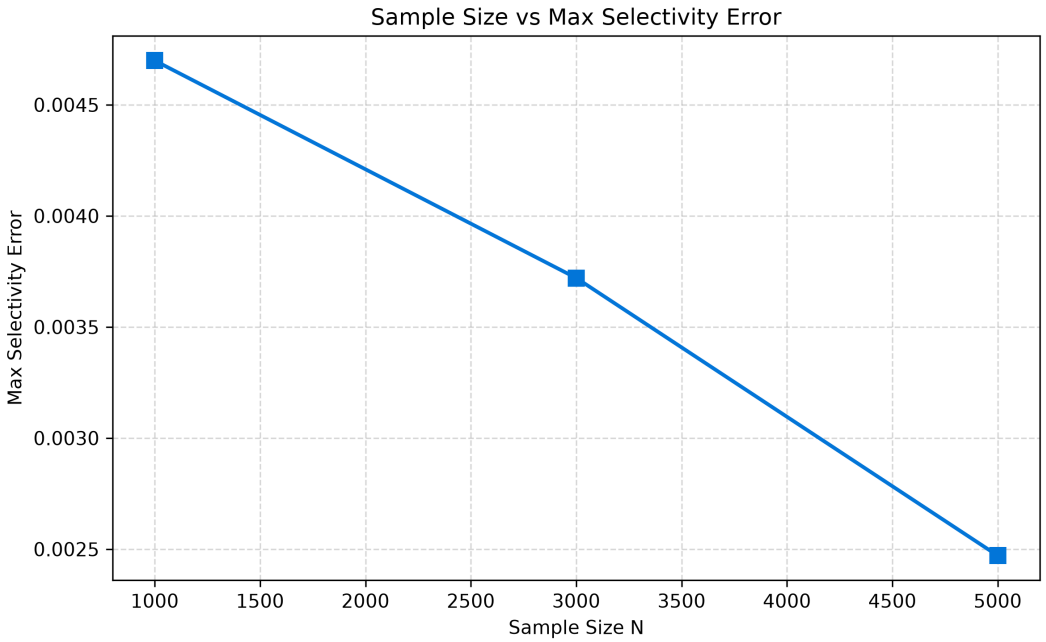


Figure 10: Sample Size vs Max Selectivity Error

3.3 9: The Trade-off of Bucket Counts

When building a histogram, choosing the number of buckets (B) is a classic trade off between performance and accuracy.

Choosing Fewer Buckets (e.g., 5 to 10)

If we use a small number of buckets, the biggest advantage is speed and efficiency. The histogram takes up very little memory in the system catalog and builds much faster because the time complexity of our algorithm,

$O(B \cdot N^2)$, drops significantly. The query planner also spends less time analyzing it. However, the major downside is that we lose a lot of data resolution. Because the buckets are so wide, the internal mathematical estimates become much less accurate, especially if the data is heavily skewed or unevenly distributed.

Choosing More Buckets (e.g., 50 to 100)

On the flip side, using a large number of buckets gives the database a high-definition view of how the data is distributed. The estimates become incredibly precise, which leads to much lower selectivity errors and helps the query optimizer make smarter decisions for complex joins. The catch is that this accuracy comes at a heavy system cost. It takes noticeably longer to calculate and construct the histogram, consumes more storage space, and adds a slight delay to the query planning phase before execution can actually begin.